

Advait Paranjpe

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Education

University of California, Los Angeles | *M.S. in Electrical and Computer Engineering* Sep 2025 – Dec 2026

- Coursework: CPU/GPU Architecture, Memory Systems, Cache Coherence, Interconnect Design.
- GPA: 3.7/4.0

The University of Manchester | *B.Eng. in Electrical and Electronic Engineering* Sep 2022 – Jul 2025

- Dissertation: “Cascaded PLL Microwave Synthesizer for 5G systems.”
- **First Class Honours** – highest UK degree classification.

Experience

Jaguar Land Rover North America | *Incoming Semiconductor Applications Engineering Intern* Jul 2026 – Sep 2026

VAST Lab, UCLA | *Student Researcher – GPU/FPGA Acceleration* Jan 2026 – Mar 2026

- Profiled ROCm attention kernels and memory bottlenecks to identify FPGA acceleration opportunities for Engram LLM inference.
- Modeled FPGA cache architectures, increasing projected hit rate from ~5% at 256 entries to ~48% with a 64k-entry, 4-way design.
- Translated workload profiling into FPGA tradeoffs across cache capacity, lookup throughput, locality, and on-chip memory usage.

NESO / National Grid ESO | *Software Engineering Intern (two summers)* Jul 2024 – Sep 2025

- Re-architected a 100k+ LOC simulation engine through vectorization and pipeline optimization, cutting runtime by 10x.
- Optimized 100M+ records through chunked, memory-efficient execution, cutting RAM from 32 GB to 16 GB and runtime by 11x.
- Reduced CPU context-switching overhead through parallel API batching and staged processing, increasing throughput by 24x for time-sensitive market telemetry.

Projects

tinyNPU | *Matrix-Multiply Accelerator with ASIC Flow (SystemVerilog)* May 2026 – Jun 2026

- Designed synthesizable RTL for a 4x4 signed-INT8 matrix-multiply accelerator with a parallel MAC array, scratchpad memories, DMA data movement, and AXI/APB wrapper variants.
- Built double-buffered memories and independent load, compute, and output FSMs, achieving 64 cycles/tile steady-state throughput.
- Improved 100 MHz WNS from -5.83 ns to -0.34 ns through STA-guided MAC pipelining, then ran an OpenLane/SKY130 RTL-to-GDS flow producing DRC/LVS-clean layouts.

Sparrow-V | *Sparse-Aware RISC-V Processor (SystemVerilog, C, Python)* Mar 2026 – Jun 2026

- Designed a synthesizable RV32I processor with a tightly coupled four-lane INT8/INT16 vector engine for quantized edge workloads.
- Implemented a custom vector ISA, pipeline control, forwarding, branch redirection, precise traps, memory backpressure, and 2:4 structured-sparse execution.
- Evaluated microarchitectural behavior through differential testing, randomized workloads, assertions, cycle counters, and synthesis-oriented area/timing analysis.

Sparrow-Cluster | *Four-Core Coherent RISC-V SoC (SystemVerilog)* Jun 2026

- Architected a four-core RV32I system with private 2 KiB L1 instruction/data caches, shared memory, and blocking cache operation.
- Designed a snoopy MSI protocol with write-back, write-allocate L1 data caches and a round-robin shared interconnect.
- Implemented invalidation/writeback handling, multicore synchronization, LR/SC, and performance counters, then validated scaling and false-sharing-sensitive workloads across 1, 2, and 4 cores.

Skills

- **Languages:** SystemVerilog, Verilog, VHDL, Tcl, Python, C, C++, Bash, SQL, MATLAB.
- **Digital Design:** Processor microarchitecture, RTL design, pipelining, cache coherence, interconnects, synthesis, STA, timing closure, and PPA analysis.
- **Tools:** Verilator, Vivado, Yosys, OpenLane, GTKWave, Git, Linux, Docker.